

Grade 11 Maths Unit 7

Unit Outcomes

By the end of this unit, students will be able to:

- **Know specific facts about types of data;**
- **Understand basic concepts about grouped data;**
- **Use statistical method to solve real life problems;**
- **Analyze statistical data;**
- **Compare statistical data; and**
- **Appreciate the value of statistics in real life.**

7.1 Types of Data

Competencies

- **identify qualitative and quantitative data;**
- **notice the difference between discrete and continuous data;**
- **identify ungrouped and grouped data;**
- **determine class interval as required to form grouped data from a given ungrouped data;**
- **make cumulative frequency table for grouped data (for both discrete and continuous data);**
- **reveal terms related to grouped continuous data, i.e., class limit, class interval and class midpoint;**
- **determine class limit, class interval and class midpoint for grouped continuous data;**
- **Organize data in group;**
- **Present data graphically;**
- **find the mean of a given grouped data;**
- **find median grouped discrete data;**
- **find Median for grouped data (continuous data);**
- **determine the quartiles for a given grouped data;**

- determine the required deciles of a given frequency distribution;
- determine the required percentile of a given frequency distribution;
- calculate interquartile range;
- determine the mode of a given grouped data; and
- identify data that are unimodal, bimodal and multimodal.

7.2 Introduction to Grouped Data

7.2.1 Grouped Discrete Data

Competencies

- identify discrete data;
- identify ungrouped and grouped data;
- determine class interval as required to form grouped data from a given ungrouped data;
- make cumulative frequency table for grouped data (for discrete data);
- reveal terms related to grouped discrete data, i.e., class limit, class interval and class midpoint;
- determine class limit, class interval and class midpoint for grouped discrete data; and
- Organize data in group.

7.2.2 Grouped Continuous Data

Competencies

- identify continuous data;
- notice the difference between discrete and continuous data;
- identify ungrouped and grouped data;
- determine class interval as required to form grouped data from a given ungrouped data;

- make cumulative frequency table for grouped data (for both discrete and continuous data);
- reveal terms related to grouped continuous data, i.e., class limit, class interval and class midpoint;
- determine class limit, class interval and class midpoint for grouped continuous data; and
- Organize data in group.

7.3 Graphical Representation of Grouped Data

Competencies

- present data graphically

7.4 Measures of Central Tendency and Their Interpretation

7.4.1 Mean for Grouped Data

Competencies

- find the mean of a given grouped data.

7.4.2 The Median of Grouped Data

Competencies

- find median grouped discrete data: and
- find Median for grouped data (continuous data)

7.4.3 Quartiles, Deciles and Percentiles

7.4.3.1 Quartiles, Deciles and Percentiles for Ungrouped Data

Competencies

- determine the quartiles for a given ungrouped data;
- determine the required deciles of a given frequency distribution; and
- determine the required percentile of a given frequency distribution.

7.4.3.2 Quartiles, Deciles and Percentiles for Grouped Data.

Competencies

- determine the quartiles for a given grouped data;
- determine the required deciles of a given frequency distribution;
- determine the required percent tile of a given frequency distribution; and
- calculate interquartile range;

7.4.4 The Mode for a Grouped Data

Competencies

- determine the mode of a given grouped data; and
- identify data that are unimodal, bimodal and multimodal.